

Task 2

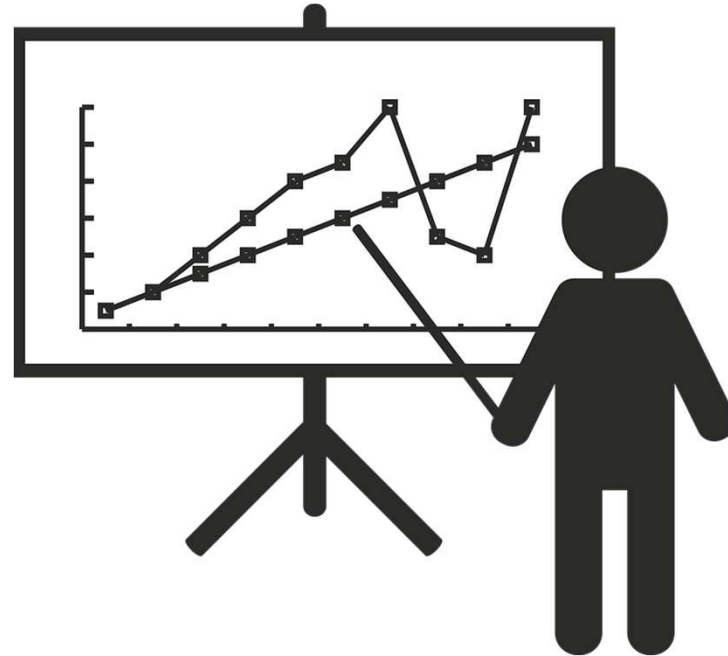
(Preview. Changes possible)

Presentation Thursday 02.07.2026
Deadline: Upload the **day before** the
lecture. If this does not work submit via
email.

Skills of a Data Scientist



Generating
Insights from Data



Communicating
Insights

Task: Consulting for EU-Park

- You are a consulting team with expertise in data science
- After your successful work for SmartBuild, your team was recommended to the (fictitious) theme park “EU-Park”
- EU-Park wants to use the potential of data science for their business and seeks your advice
- Your target group is the board of EU-Park (CEO + CTO)



Image created by AI

Motivation and background

- EU-Park has data on
 - a) Wait times for their attractions
 - b) Menu purchases at their restaurants
 - c) Customer data on the purchase of season passes (“gold pass” and “silver pass”). The data reflects the information about customers at the first of January from last year. On this date, they had to decide if they buy a season pass or not.

All data is provided on Felix in the folder “Task_2”

Motivation and background

- EU-Park wants to ...
 - a) ... predict wait times and understand what impacts wait times (in other words, what variables allow for a good/reasonable prediction)
 - b) ... offer special discounts on certain menu combinations to “gold pass” owners. Therefore EU-Park wants to understand if there are certain groups of food items that are popular in combination. In other words, they wonder if there is a group structure in the purchase behavior that could drive the special offer for menu combinations.
 - c) ... predict which customer will purchase season passes (“gold pass” and “silver pass”) and understand what drives the decision. (in other words, what variables allow for a good/reasonable prediction)

The Task

- Address the goals of EU park from the previous slide.
- Present your findings to the CEO and CTO (fictitious) and prepare a corresponding presentation.
- Make at least one analysis per group member, so that you have proper content for the presentation time.
- At least one area of interest (a, b, c from previous slide) per group member must be addressed. (Groups bigger than 3 must make deeper analysis of one area. Groups smaller than 3 do not need to address all areas.)

Grading

- **Assignments during the term (50%)**
 - Technical quality (Code and results) 50%
 - Presentation (How you present and discuss results) 50%
- **Exam (50%)**
 - Question about theoretical foundation
 - Practical implementation and related questions (e.g. complete, interpret or rewrite some code)

Presentation times

- Presentation times
 - <3 persons: 10 min
 - 3 persons: 12 min
 - 4 persons : 15min
 - 5 persons: 19 min
 - 6 persons: 23 min
- **Timing is strict! You will not have more time. If time runs out, you need to stop. No extra time.**
- (Somewhat odd numbers due to time constraints.)
- Groups will start in reverse order from the first presentation.

Technology to use

- The task is to be implemented with Python and optionally other tools (e.g. for additional visualization).
- The majority of the work is to be implemented with Python.

Use of LLMs (e.g. ChatGPT)

- For this course the following applies until further notice:
 - LLMs are treated like other existing programming tools or help pages.
 - You must be able to explain your solution in detail (i.e. you must understand what you are doing or have done with the help of LLMs/ChatGPT).
 - Presentation discussion